

Backstepping Sliding Mode Control for Trajectory Tracking of a Differential-Drive Wheeled Mobile Robot via ESP-NOW Wireless Communication

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Abstract—This paper presents the implementation and experimental validation framework of a velocity-level Backstepping Sliding Mode Controller (BSMC) for trajectory tracking of a differential-drive wheeled mobile robot (WMR). The controller combines a Kanayama-form backstepping tracking law with two coupled sliding surfaces and a boundary-layer saturation function to improve robustness against practical disturbances such as actuator imperfections, sensor noise, encoder quantization, wheel asymmetry, and wireless timing variation. The control architecture is realized on a low-cost distributed platform consisting of an STM32F411 microcontroller, an ESP32 wireless bridge using the ESP-NOW protocol, and a ROS2 supervisor running on a host PC. The controller operates at the velocity level: it computes body-frame velocity commands rather than wheel torques directly, and these commands are tracked by inner-loop PID wheel-speed controllers on the STM32. An overhead camera with AprilTag markers is integrated as an external pose-observation module for experimental validation and future localization correction. The paper describes the system architecture, velocity-level modeling, controller implementation, wireless packet structure, timestamp handling, and evaluation metrics for comparing Backstepping-only and BSMC on circular and figure-eight trajectories.

Index Terms—Backstepping sliding mode control; differential-drive wheeled mobile robot; nonholonomic systems; ESP-NOW wireless telemetry; AprilTag localization; trajectory tracking.

I. INTRODUCTION

Trajectory tracking of wheeled mobile robots (WMRs) is a fundamental problem in mobile robotics, with applications ranging from warehouse automation to indoor service robots. For differential-drive robots, the nonholonomic constraint makes the tracking problem nontrivial. In practical hardware platforms, the tracking performance is further affected by limited computational resources, wireless communication latency and packet loss, PWM deadzone nonlinearities, wheel asymmetry, floor friction, encoder quantization, and inertial-sensor noise.

Backstepping control provides a systematic, constructive procedure for stabilizing nonholonomic mobile robots [1].

Sliding Mode Control (SMC) augments this with robustness against bounded matched disturbances [2], [4]. The combination—Backstepping Sliding Mode Control (BSMC)—has been studied for trajectory tracking of WMRs in recent works [3]. However, many existing works emphasize theoretical controller design or simulation, while fewer papers discuss the full implementation loop on a low-cost physical WMR, including wireless telemetry and embedded inner-loop wheel-speed control.

Indoor localization and pose estimation are also critical in mobile robot experiments. Wheel odometry provides high-rate relative motion information, but it accumulates drift due to wheel slip, floor unevenness, and parameter uncertainty. To reduce localization error, multi-sensor fusion and Kalman-filter approaches using odometry, IMU, UWB, LiDAR, or other absolute measurements have been investigated [6]–[9]. AprilTag-based localization provides a low-cost visual pose source for indoor mobile robots when camera calibration and marker geometry are known [10]–[12]. In this paper, the current closed-loop tracking experiment relies on dead-reckoning feedback, while AprilTag detection is used as an external pose-observation module for validation and future localization correction.

Low-cost wireless communication is another practical issue in embedded mobile robot systems. ESP-NOW has been investigated for indoor performance, energy efficiency, and low-cost robotics applications [13]–[16]. Its peer-to-peer operation is suitable for exchanging compact velocity commands and telemetry packets between a ROS2 host and an embedded controller. Nevertheless, indoor reflections, obstacles, and packet timing variation must be considered when ESP-NOW is used inside a closed-loop robotic platform.

Motivated by these issues, this paper develops a low-cost ROS2–ESP-NOW–STM32 implementation of a velocity-level BSMC controller for a differential-drive WMR. Unlike studies that focus only on controller equations, this work emphasizes the full implementation chain: high-level control

in ROS 2, wireless command and telemetry exchange through ESP-NOW, inner-loop wheel-speed PID control on STM32, and external pose observation using AprilTag detection.

The contributions of this paper are as follows:

- 1) An **experimental implementation and validation framework** of a velocity-level BSMC trajectory tracking controller for a differential-drive WMR, with comparison against a Backstepping-only baseline.
- 2) A **low-cost embedded and wireless architecture** using an STM32F411 microcontroller, an ESP32/ESP-NOW wireless bridge, and ROS 2, together with an overhead AprilTag camera for external pose observation.
- 3) A **velocity-level actuator approximation** and practical implementation details, including PWM deadzone compensation, wheel asymmetry calibration, encoder filtering, packet structure, and timestamp handling.
- 4) A **quantitative evaluation protocol** based on trajectory plots, position-error norm, heading error, RMSE, MAE, maximum error, convergence time, and ESP-NOW link-quality metrics.

II. SYSTEM MODELING

A. Kinematic Model

The pose of the WMR is $q = [x, y, \theta]^\top$ in the world frame. The nonholonomic kinematic model is

$$\dot{x} = v \cos \theta, \quad \dot{y} = v \sin \theta, \quad \dot{\theta} = \omega, \quad (1)$$

where v is the linear velocity and ω is the angular velocity. This model is commonly used for differential-drive mobile robots [5].

B. Differential-Drive Forward Kinematics

Wheel rim speeds are mapped to body velocities by

$$v = \frac{v_R + v_L}{2}, \quad \omega = \frac{v_R - v_L}{L}, \quad (2)$$

where v_L, v_R are the left/right wheel rim speeds and $L = 0.17$ m is the wheelbase.

C. Actuator Dynamics and Velocity-Level Approximation

The STM32 receives velocity commands v_{cmd} and ω_{cmd} , converts them to wheel-speed references via inverse kinematics, and closes PID loops locally. The resulting velocity-level actuator model is

$$\dot{v} = k_v(v_{\text{cmd}} - v) + \Delta_v(t), \quad (3)$$

$$\dot{\omega} = k_\omega(\omega_{\text{cmd}} - \omega) + \Delta_\omega(t), \quad (4)$$

where the lumped disturbances Δ_v, Δ_ω encompass Coulomb and viscous friction, PWM deadzone ($\approx 12\%$), wheel asymmetry, and encoder quantization noise.

Assumption 1 (bounded disturbances): $|\Delta_v| \leq \bar{\Delta}_v, |\Delta_\omega| \leq \bar{\Delta}_\omega$.

Assumption 2 (bounded inner-loop error): $|v - v_{\text{cmd}}| \leq \varepsilon_v, |\omega - \omega_{\text{cmd}}| \leq \varepsilon_\omega$.

TABLE I
BSMC CONTROLLER PARAMETERS

Parameter	Symbol	Circle	Figure-8
Longitudinal gain	k_1	0.8	0.8
Lateral gain	k_2	2.4	2.4
Heading gain	k_3	4.0	4.0
Sliding coupling	c	1.0	1.0
SMC gain 1	K_{s1}	0.001	0.002
SMC gain 2	K_{s2}	0.002	0.005
Boundary-layer 1	φ_1	0.80	0.45
Boundary-layer 2	φ_2	1.50	1.20
Max linear vel.	v_{max}	0.35 m/s	
Max angular vel.	ω_{max}	0.6 rad/s	

III. BSMC CONTROL LAW AND STABILITY ANALYSIS

A. Tracking Error in the Robot Frame

Let the body-frame tracking error be obtained from the world-frame position/heading error through the standard rotation transform $R(\theta)$:

$$e_x = (x_d - x) \cos \theta + (y_d - y) \sin \theta, \quad (5)$$

$$e_y = -(x_d - x) \sin \theta + (y_d - y) \cos \theta, \quad (6)$$

$$e_\theta = \theta_d - \theta. \quad (7)$$

The wrapped definition of e_θ avoids discontinuities when the heading error crosses $\pm\pi$. The corresponding error dynamics are

$$\dot{e}_x = \omega e_y - v + v_d \cos e_\theta, \quad (8)$$

$$\dot{e}_y = -\omega e_x + v_d \sin e_\theta, \quad (9)$$

$$\dot{e}_\theta = \omega_d - \omega. \quad (10)$$

B. Sliding Surfaces

Two coupled sliding surfaces are defined as

$$s_1 = e_x, \quad s_2 = e_\theta + c e_y, \quad c > 0. \quad (11)$$

C. BSMC Control Law

$$v_{\text{cmd}} = v_d \cos e_\theta + k_1 e_x + K_{s1} \text{sat}(s_1/\varphi_1), \quad (12)$$

$$\omega_{\text{cmd}} = \omega_d + v_d(k_2 e_y + k_3 \sin e_\theta) + K_{s2} \text{sat}(s_2/\varphi_2), \quad (13)$$

where $v_d(t)$ and $\omega_d(t)$ are the desired velocity profiles from the trajectory generator, and

$$\text{sat}(z) = \max(-1, \min(1, z)), \quad z = s/\varphi, \quad (14)$$

provides chattering mitigation via boundary-layer regularization.

Controller gains for both trajectories are listed in Table I.

D. Stability Analysis

Theorem: Under Assumptions 1–2, the closed-loop system converges to a *Uniform Ultimate Bound (UUB)* around the origin of (e_x, e_y, e_θ) .

Proof sketch: Choose the Lyapunov candidate $V = \frac{1}{2}s_1^2 + \frac{1}{2}s_2^2$. Within the boundary layer ($|s_i| \leq \varphi_i$), the saturation function acts as a linear damping term; outside it, $\dot{V} < 0$ for

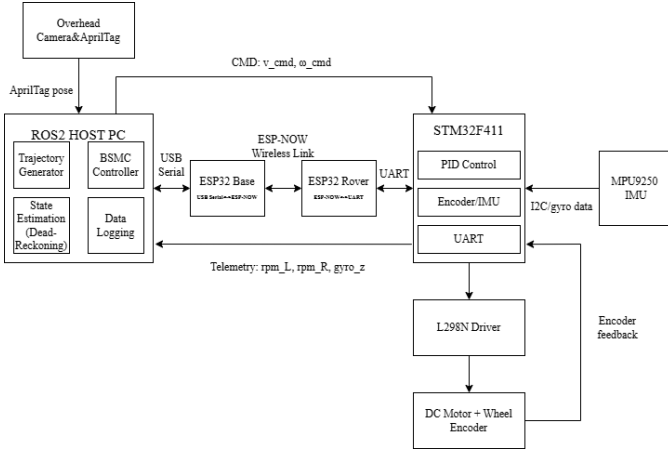


Fig. 1. Hardware and communication architecture of the proposed BSMC platform.

TABLE II
HARDWARE PARAMETERS

Parameter	Symbol	Value	Source
Wheel diameter	D	0.063 m	Datasheet
Wheelbase	L	0.17 m	Measured
Encoder PPR	—	937	Firmware
Control period	T_s	0.05 s (20 Hz)	TIM10
PWM deadzone	PWM_{dead}	$\approx 12\%$	MOTOR_PWM_MIN
Wheel calib. factor	K_{calib}	1.22	Firmware
Encoder LPF	α_{fw}	0.30	Firmware
Linear bandwidth	k_v	[TBD – step test]	Experiment
Angular bandwidth	k_ω	[TBD – step test]	Experiment
Robot mass	m	[TBD – weigh]	Experiment

sufficiently large K_{s1} and K_{s2} relative to $\bar{\Delta}_v$ and $\bar{\Delta}_\omega$. Under ideal conditions ($\varepsilon_v = \varepsilon_\omega = 0$ and $\bar{\Delta} = 0$), the bound tightens to asymptotic convergence. ■

IV. HARDWARE ARCHITECTURE

A. System Overview

The system comprises three layers: an STM32F411 motor-control board, an ESP32 ESP-NOW wireless bridge pair, and a ROS2 host PC that receives absolute pose corrections from an overhead AprilTag camera, as illustrated in Fig. 1.

B. STM32F411 Firmware

The STM32 runs a 20 Hz control loop (TIM10 interrupt, $T_s = 0.05$ s). Key hardware parameters are listed in Table II. Each wheel is regulated by an independent PID speed controller. A first-order EMA filter ($\alpha_{fw} = 0.30$) is applied to the raw encoder RPM. PWM deadzone compensation adds a fixed offset (MOTOR_PWM_MIN = 120/999 $\approx 12\%$) for any nonzero command, and left-wheel asymmetry is corrected by a calibration factor $K_{calib} = 1.22$.

C. ESP-NOW Wireless Link

A peer-registered IEEE 802.11 ESP-NOW link connects the on-robot ESP32 to the PC-side ESP32. Two packet types

are used: CMD (ROS2 $\rightarrow$ robot) and DATA (robot $\rightarrow$ ROS2). The CMD packet carries the host timestamp, sequence number, v_{cmd} , and ω_{cmd} . The DATA packet carries the robot timestamp, sequence number, rpm_L , rpm_R , and $gyro_z$. The nominal communication rate is 20 Hz. Packet timestamps and sequence numbers are logged for latency, jitter, and packet-loss evaluation. If the received odometry interval exceeds a predefined threshold, the `state_bridge` node clamps the integration step to avoid sudden odometry jumps caused by delayed packets.

D. ROS2 State Estimation

The `state_bridge.py` node implements dead-reckoning with midpoint integration:

$$\theta_{k+1} = \theta_k + \omega_{enc} \Delta t, \quad (15)$$

$$x_{k+1} = x_k + v \cos\left(\frac{\theta_k + \theta_{k+1}}{2}\right) \Delta t, \quad (16)$$

$$y_{k+1} = y_k + v \sin\left(\frac{\theta_k + \theta_{k+1}}{2}\right) \Delta t. \quad (17)$$

A first-order LPF ($\alpha = 0.60$) smooths the velocity estimate. This is dead-reckoning only; an EKF fusing odometry, IMU, and AprilTag corrections is planned as future work.

E. AprilTag External Pose Observation

An overhead IP camera streams RTSP video at 30 Hz. The `camera.py` node detects tag36h11 markers (tag size 0.150 m, tag ID 0) and estimates pose via `cv2.solvePnP` using calibrated camera intrinsics. Camera intrinsics are $f_x = 767.68$, $f_y = 765.51$, $c_x = 637.44$, and $c_y = 357.26$. In the current experiment, AprilTag pose is used as an external pose-observation source for validation and future localization correction. It is not fused into the closed-loop controller.

V. EXPERIMENTAL METHODOLOGY

A. Actuator Bandwidth Identification

Linear step test: command $v_{cmd} : 0 \rightarrow 0.15$ m/s while holding $\omega_{cmd} = 0$. Measure the time required to reach 63.2 % of the steady-state value from `/odom_raw`; then compute $k_v = 1/T_v$.

Angular step test: command $\omega_{cmd} : 0 \rightarrow 0.3$ rad/s while holding $v_{cmd} = 0$. The angular bandwidth is computed as $k_\omega = 1/T_\omega$.

B. Trajectory Definitions

Circle: radius $R = 1.0$ m, angular speed $\Omega = 0.25$ rad/s, nominal $v_d = R\Omega = 0.125$ m/s, with a 2 s linear ramp:

$$v_d(t) = V_D \cdot t/T_{ramp}, \quad \omega_d(t) = \Omega \cdot t/T_{ramp}. \quad (18)$$

Figure-eight: Lissajous curve $x = A \sin(\Omega t)$, $y = B \sin(2\Omega t)$ with $A = 0.5$ m, $B = 0.25$ m, and $\Omega = 0.10$ rad/s; nominal speed:

$$v_d = \sqrt{(A\Omega)^2 + (2B\Omega)^2}. \quad (19)$$

TABLE III
TRAJECTORY TRACKING PERFORMANCE (RMSE)

Controller	Traj.	RMSE e_x (m)	RMSE e_y (m)	RMSE e_θ (rad)	Max err. (m)	Conv. (s)
Backstepping	Circle	—	—	—	—	—
BSMC	Circle	—	—	—	—	—
Backstepping	Figure-8	—	—	—	—	—
BSMC	Figure-8	—	—	—	—	—

[Fig. 2: XY trajectory – Circle.
Reference path, Backstepping-only, BSMC, and AprilTag ground truth.]

Fig. 2. Circle-trajectory tracking ($R = 1$ m, $v_d = 0.125$ m/s): reference path, Backstepping-only, and BSMC, measured via wheel odometry and AprilTag camera.

[Fig. 3: XY trajectory – Figure-eight.
Reference path, Backstepping-only, BSMC, and AprilTag ground truth.]

Fig. 3. Figure-eight trajectory tracking ($A = 0.5$ m, $B = 0.25$ m, $\Omega = 0.10$ rad/s): reference path, Backstepping-only, and BSMC.

[Fig. 4: Error time series.
 $e_x(t)$, $e_y(t)$, and $e_\theta(t)$ for Backstepping-only vs. BSMC.]

Fig. 4. Tracking-error time series (e_x , e_y , e_θ) for Backstepping-only and BSMC on the circle trajectory.

TABLE IV
ESP-NOW LINK QUALITY

Condition	Dist.	Median lat.	Jitter (95 %)	Loss
Static	5 m	—	—	—
Moving	5 m	—	—	—
Static	10 m	—	—	—
Static	20 m	—	—	—

C. Baseline Comparison

Two controllers are compared: (i) *Backstepping-only* ($K_{s1} = K_{s2} = 0$, no sliding terms) and (ii) *BSMC* ($K_{s1}, K_{s2} > 0$, with boundary-layer saturation). Both controllers are evaluated under the same trajectory parameters, initial pose, and experimental setup. At least three runs are performed per scenario, and mean $\pm$ std values are reported.

D. Evaluation Metrics

The position-error norm is defined as

$$e_p(k) = \sqrt{(x_d(k) - x(k))^2 + (y_d(k) - y(k))^2}. \quad (20)$$

The heading error is wrapped to $(-\pi, \pi]$ by

$$e_\theta(k) = \text{atan2}(\sin(\theta_d(k) - \theta(k)), \cos(\theta_d(k) - \theta(k))). \quad (21)$$

The position RMSE is computed as

$$\text{RMSE}_p = \sqrt{\frac{1}{N} \sum_{k=1}^N e_p^2(k)}. \quad (22)$$

The component-wise RMSE values are

$$\text{RMSE}_{e_i} = \sqrt{\frac{1}{N} \sum_{k=1}^N e_i^2(k)}, \quad e_i \in \{e_x, e_y, e_\theta\}. \quad (23)$$

The mean absolute position error and maximum position error are

$$\text{MAE}_p = \frac{1}{N} \sum_{k=1}^N |e_p(k)|, \quad (24)$$

$$e_{p,\max} = \max_{1 \leq k \leq N} e_p(k). \quad (25)$$

The convergence time is defined as the first time instant after which e_p remains below a threshold ϵ_p for a continuous duration T_{hold} :

$$T_{\text{conv}} = \min\{t_k : e_p(t) \leq \epsilon_p, \forall t \in [t_k, t_k + T_{\text{hold}}]\}. \quad (26)$$

In the experiments, ϵ_p and T_{hold} are set to [TBD] and [TBD], respectively.

E. Ground-Truth Sources

Three levels of ground truth are used: (1) wheel odometry with midpoint integration and LPF; (2) AprilTag camera at 30 Hz; and (3) pen trace on paper for qualitative evaluation.

VI. RESULTS

A. Tracking Performance

Fig. 2 and Fig. 3 show the XY trajectories for both controllers on the circle and figure-eight paths, respectively. Fig. 4 shows the tracking-error time series. Quantitative results are summarized in Table III.

B. ESP-NOW Link Quality

Table IV reports link latency and packet-loss rate measured at three distances under static and moving conditions.

VII. CONCLUSION AND FUTURE WORK

This paper presented the implementation framework of a velocity-level BSMC for trajectory tracking of a differential-drive WMR, implemented on an STM32F411 + ESP32/ESP-NOW + ROS2 platform. The controller augments the Kanayama backstepping framework with two coupled sliding surfaces and a boundary-layer saturation function. Practical implementation issues, including PWM deadzone ($\approx 12\%$), wheel asymmetry ($K_{\text{calib}} = 1.22$), encoder LPF ($\alpha = 0.30$), packet structure, timestamp handling, and AprilTag-based external pose observation, were described.

The current work focuses on trajectory tracking with dead-reckoning feedback and external AprilTag validation. The main limitations are odometry drift during long-duration experiments, sensitivity to wheel slip and uneven floor conditions, encoder quantization, IMU yaw-rate noise, and possible packet timing variation in the ESP-NOW link. The system has not yet addressed obstacle avoidance, global path planning, or fully fused localization in the closed loop.

Planned extensions include: (i) EKF fusion of wheel odometry, IMU gyroscope, and AprilTag pose for improved state estimation; (ii) extension to leader-follower multi-robot formation control with NDO-based adaptive super-twisting SMC; (iii) more extensive ESP-NOW latency and packet-loss evaluation; and (iv) adaptive online tuning of BSMC gains and boundary-layer widths.

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